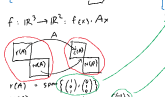
5+3

SVD:

$$A = U\Sigma V^T$$

$$\begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix} = \underbrace{\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}}_{U} \underbrace{\begin{bmatrix} \sqrt{2} & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}}_{\Sigma} \underbrace{\begin{bmatrix} 1/\sqrt{2} & 0 & -1/\sqrt{2} \\ 0 & 1 & 0 \\ 1/\sqrt{2} & 0 & 1/\sqrt{2} \end{bmatrix}^T}_{V}$$

$f: \mathbb{R}^3 \to \mathbb{R}^2 : f(x) = Ax$



$x = \begin{pmatrix} \sqrt{2} \\ 1 \\ 2\sqrt{2} \end{pmatrix}$

$r(A) = \text{span}\left\{ \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \end{pmatrix} \right\}$

$N(A) = N\begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix} = \text{span}\left\{ \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix} \right\}$

$C(A) = \mathbb{R}^2$

$N(A^T) = \{0\}$

$$\begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix} x = \underbrace{\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}}_{U} \underbrace{\begin{bmatrix} \sqrt{2} & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}}_{\Sigma} \underbrace{\begin{bmatrix} 1/\sqrt{2} & 0 & -1/\sqrt{2} \\ 0 & 1 & 0 \\ 1/\sqrt{2} & 0 & 1/\sqrt{2} \end{bmatrix}^T}_{V} x$$

$x = \begin{pmatrix} \sqrt{2} \\ 1 \\ 2\sqrt{2} \end{pmatrix}$

$$\begin{bmatrix} 1/\sqrt{2} & 0 & 1/\sqrt{2} \\ 0 & 1 & 0 \\ -1/\sqrt{2} & 0 & 1/\sqrt{2} \end{bmatrix} \begin{pmatrix} \sqrt{2} \\ 1 \\ 2\sqrt{2} \end{pmatrix} =$$

$$\begin{bmatrix} \sqrt{2} & 0 \\ 0 & 1 \end{bmatrix} \begin{pmatrix} 3 \\ 1 \end{pmatrix} = \begin{pmatrix} 3\sqrt{2} \\ 1 \end{pmatrix}$$

$$\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{pmatrix} 3\sqrt{2} \\ 1 \end{pmatrix} = \begin{pmatrix} 1 \\ 3\sqrt{2} \end{pmatrix}$$

$$\begin{pmatrix} \begin{pmatrix} 1/\sqrt{2} \\ 0 \\ 1/\sqrt{2} \end{pmatrix} \cdot \begin{pmatrix} \sqrt{2} \\ 1 \\ 2\sqrt{2} \end{pmatrix} \\ \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} \cdot \begin{pmatrix} \sqrt{2} \\ 1 \\ 2\sqrt{2} \end{pmatrix} \\ \begin{pmatrix} -1/\sqrt{2} \\ 0 \\ 1/\sqrt{2} \end{pmatrix} \cdot \begin{pmatrix} \sqrt{2} \\ 1 \\ 2\sqrt{2} \end{pmatrix} \end{pmatrix} = \begin{pmatrix} 3 \\ 1 \\ -1 \end{pmatrix}$$

$$\begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix} \begin{pmatrix} \sqrt{2} \\ 1 \\ 2\sqrt{2} \end{pmatrix} = \begin{pmatrix} 1 \\ 3\sqrt{2} \end{pmatrix}$$

$$A = U\Sigma V^T$$

$$A^TA = V\Sigma^TU^TU\Sigma V^T$$

$$= V\Sigma^T\Sigma V^T$$

$$A = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix} \qquad A^TA = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}$$

$$A^T = \begin{bmatrix} 0 & 1 \\ 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$\begin{vmatrix} 1-\lambda & 0 & 1 \\ 0 & 1-\lambda & 0 \\ 1 & 0 & 1-\lambda \end{vmatrix} = 0$$

$$(1-\lambda)^3 - (1-\lambda)$$

$$= (1-\lambda)\left((1-\lambda)^2 - 1\right)$$

$$= (1-\lambda)(\lambda^2 - 2\lambda) + \text{[illegible]}$$

$$= (1-\lambda)(\lambda-2)\lambda = 0$$

$$\lambda = 2, 1, 0$$

For $\lambda = 2$

$$N\begin{bmatrix} 1-2 & 0 & 1 \\ 0 & 1-2 & 0 \\ 1 & 0 & 1-2 \end{bmatrix} = N\begin{bmatrix} -1 & 0 & 1 \\ 0 & -1 & 0 \\ 1 & 0 & -1 \end{bmatrix}$$

$$N\begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & 0 \end{bmatrix}$$

$$E_2 = \text{span}\left\{ \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} \right\}$$

For $\lambda = 1$

$$N\begin{bmatrix} 1-1 & 0 & 1 \\ 0 & 1-1 & 0 \\ 1 & 0 & 1-1 \end{bmatrix} = N\begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$E_1 = \text{span}\left\{ \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} \right\}$$

For $\lambda = 0$

$$N\begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix} = N\begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$$

$$E_0 = \text{span}\left\{ \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix} \right\}$$

$$V = \begin{bmatrix} 1/\sqrt{2} & 0 & -1/\sqrt{2} \\ 0 & 1 & 0 \\ 1/\sqrt{2} & 0 & 1/\sqrt{2} \end{bmatrix}$$

$$\Sigma = \begin{bmatrix} \sigma_1 & 0 & 0 \\ 0 & \sigma_2 & 0 \end{bmatrix} = \begin{bmatrix} \sqrt{2} & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}$$

$$\Sigma^T\Sigma = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$A = U\Sigma V^T \qquad AV = U\Sigma V^TV$$

$$AV = U\Sigma$$

$$Av_1 = u_1\sigma_1$$

$$\frac{Av_1}{\sigma_1} = u_1 \qquad \frac{Av_2}{\sigma_2} = u_2$$

$$u_1 = \frac{1}{\sqrt{2}}\begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}\begin{pmatrix} 1/\sqrt{2} \\ 0 \\ 1/\sqrt{2} \end{pmatrix} = \frac{1}{\sqrt{2}}\begin{pmatrix} 0 \\ \sqrt{2} \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

$$u_2 = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}\begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

$$U = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

$$A = U\Sigma V^T$$

$$A^TA = V\Sigma^TU^TU\Sigma V^T$$

$$= V\Sigma^T\Sigma V^T$$

$$(A^TA)^T = A^TA \quad \therefore A \text{ symmetric}$$

$$\Sigma^T\Sigma = \begin{bmatrix} \sigma_1^2 & 0 & 0 \\ 0 & \sigma_2^2 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\Sigma = \begin{bmatrix} \sigma_1 & 0 & 0 \\ 0 & \sigma_2 & 0 \end{bmatrix} \qquad \sigma_1 = \sqrt{2}, \quad \sigma_2 = \sqrt{2}$$

$$A^TAx = \lambda x$$

$$x^TA^TAx = x^T\lambda x$$

$$(Ax)^T(Ax) = \lambda(x^Tx)$$

$$|Ax|^2 = \lambda|x|^2$$

$$\frac{|Ax|^2}{|x|^2} = \lambda$$

$$AA^T = U\Sigma V^TV\Sigma^TU^T$$

$$= U\Sigma\Sigma^TU^T$$

$$\begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}\begin{bmatrix} 0 & 1 \\ 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix}$$

$$\lambda = 1, 2$$

$$E_1 = \text{span}\left\{ \begin{pmatrix} 1 \\ 0 \end{pmatrix} \right\}$$

$$E_2 = \text{span}\left\{ \begin{pmatrix} 0 \\ 1 \end{pmatrix} \right\}$$

$$U = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

$$\begin{bmatrix} \sigma_1 & 0 & 0 \\ 0 & \sigma_2 & 0 \end{bmatrix}\begin{bmatrix} \sigma_1 & 0 \\ 0 & \sigma_2 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} \sigma_1^2 & 0 \\ 0 & \sigma_2^2 \end{bmatrix} = \begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix}$$

$$\sigma_1 = \sqrt{2}, \quad \sigma_2 = 1$$

$$\Sigma = \begin{bmatrix} \sqrt{2} & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}$$

$$A = U\Sigma V^T$$

$$AV = U\Sigma \qquad Av_1 = \sigma_1 u_1$$

$$A^TU = V\Sigma^TU^TU\Sigma$$

$$A^TU = V\Sigma^T \qquad A^Tu_1 = \sigma_1 v_1$$

$$v_1 = \frac{A^Tu_1}{\sigma_1} \qquad v_2 = \frac{A^Tu_2}{\sigma_2}$$

$$v_1 = \frac{1}{\sqrt{2}}\begin{bmatrix} 0 & 1 \\ 1 & 0 \\ 0 & 1 \end{bmatrix}\begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 1/\sqrt{2} \\ 0 \\ 1/\sqrt{2} \end{pmatrix}$$

$$V = \begin{bmatrix} 1/\sqrt{2} & 0 & -1/\sqrt{2} \\ 0 & 1 & 0 \\ 1/\sqrt{2} & 0 & 1/\sqrt{2} \end{bmatrix}$$

$$\begin{pmatrix} 1/\sqrt{2} \\ 0 \\ -1/\sqrt{2} \end{pmatrix}$$

$$v_2 = \begin{bmatrix} 0 & 1 \\ 1 & 0 \\ 0 & 1 \end{bmatrix}\begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}$$

$$N\begin{bmatrix} 1/\sqrt{2} & 0 & 1/\sqrt{2} \\ 0 & 1 & 0 \end{bmatrix} = N\begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$$

$$= \text{span}\left\{ \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix} \right\}$$